rapid growth, and helped them displace other leaders. Both firms leveraged microservices-driven technology with containers and serverless computing.

Let's look at containers and serverless computing in detail, with their pros, cons, and best practices.

Containers

Containers are a type of virtualisation technology that allows you to run multiple isolated applications on the same operating system instance. They provide an efficient way to run applications in different environments, making them ideal for testing and development as well as production deployments. Containers provide a lightweight, easy-to-use alternative to traditional virtual machines. They are becoming increasingly popular due to their portability and flexibility, allowing developers to easily move applications from one environment to the next without worrying about compatibility issues. With containers, you can easily manage complex, multi-service applications in an isolated and consistent environment. Plus, they're perfect for testing and development environments.

Serverless computing

Serverless is a cloud computing model where the cloud provider manages the infrastructure and automatically allocates resources as necessary. When using serverless, developers only need to worry about writing code, with the cloud provider handling everything else, including scaling, maintenance, and security. Serverless cloud technologies are best suited for event-based applications and services that don't require persistent running instances. They are particularly useful for applications that need to scale quickly or experience unpredictable usage patterns. By unleashing the power of event-driven computing with serverless, you can encapsulate and scale your services without breaking a sweat. It's the ideal choice for unpredictable usage patterns or applications that require ultra-fast response times.

Since both containers and serverless computing do not need to be associated with hardware or underlying operating systems, they do have some common benefits. Table 1 lists a few of these.

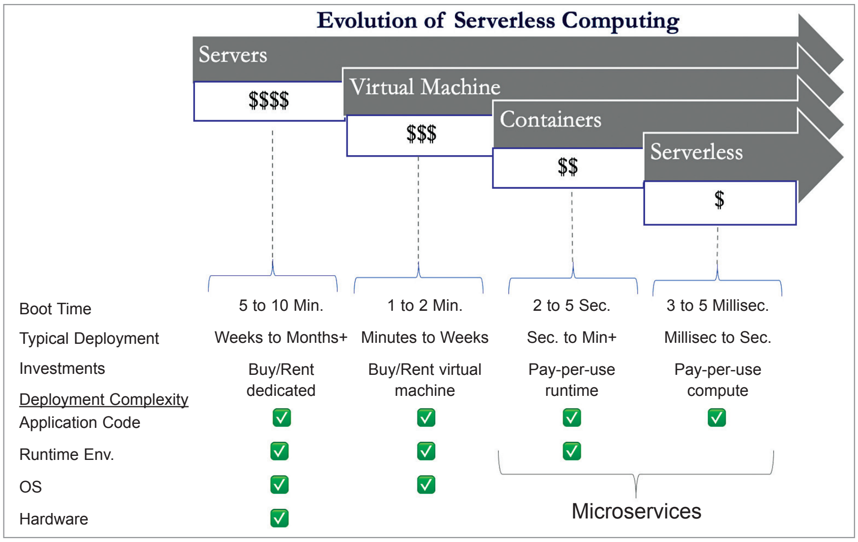

Fig. 1: Evolution of application design and deployment to serverless computing

TABLE 1
CONTAINERS AND SERVERLESS COMPUTING: A COMPARISON

Testing and development. Containers provide a safe and isolated environment for developers to test and run code without impacting other projects or environments.	***Backend as a Service (BaaS).*** Serverless is ideal for building backend services, such as REST APIs, that require minimal configuration and can scale automatically.
Microservices. Containers make it easy to separate application functionality into smaller components that can be scaled and deployed independently.	***Event-driven apps.*** Serverless is well-suited for applications that have specific events that drive their behaviour, such as chatbots or IoT devices. It has also become popular in tech modernisation.
Hybrid cloud environments. Containers offer a consistent environment across different cloud providers and on-premise resources, making them ideal for hybrid cloud environments.	***Batch processing.*** Serverless is a great option for processing large amounts of data in batch, without having to worry about the underlying infrastructure.

CLOUD: THE KEY TRENDS

Let's look at the key advancements in cloud technology as a result of the use of containers and serverless computing.

Software as a Service (SaaS)

SaaS is a cloud-based software delivery model that provides access to applications over the internet. SaaS applications are typically subscription based and are hosted by the provider. It has a number of benefits, including:

Speed. SaaS applications can be deployed quickly and easily.

Cost. These applications are typically more cost-effective than traditional on-premises applications.

Scalability. SaaS applications can be easily scaled up or down to meet demand.